

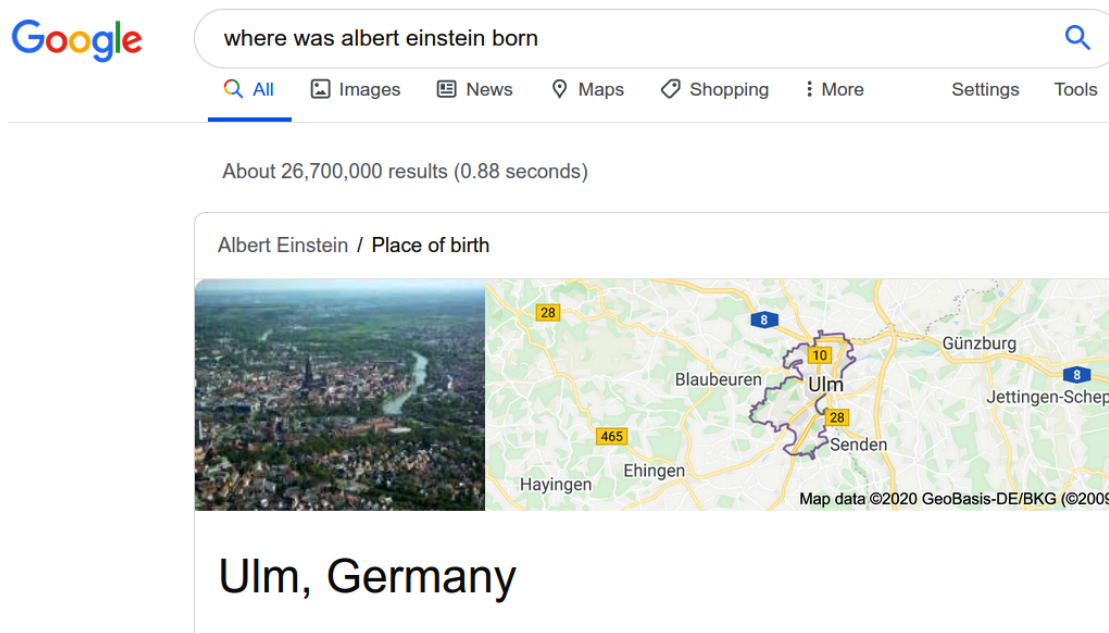
11.1 Named Entity Recognition

Find people, places, organizations, and similar spans.

These notes walk through the lecture slides. The original deck is unchanged; use **(slides)** on the course hub if you want that PDF.

1. What NER is

A user types "Where was Albert Einstein born?" The search engine cannot answer *Ulm, Germany* unless it first sees that **Albert Einstein is a person**. That step is named entity recognition (NER).



NER finds **spans** in a document and labels each span with a type: person, location, organization, money, date, product, law, and whatever else your domain needs. It is not the same as keyword search. The string *Apple* is a company in one sentence and a fruit in another.

2. Why it sits in so many pipelines

NER is the IE step that later tasks wait on. Relation extraction and event extraction need entities first. Machine translation often **leaves names untranslated**. A chatbot fills slots from the same spans.

displaCy Named Entity Visualizer

Justin Pierre James Trudeau PC MP (born December 25, 1971) is a Canadian politician serving as the 23rd prime minister of Canada since 2015 and Leader of the Liberal Party since 2013. Trudeau is the second-youngest Canadian Prime Minister after Joe Clark; he is also the first to be related to a previous holder of the post, as the eldest son of Pierre Trudeau.

Model

English - en_core_web_lg (v2.0.0)

Entity labels (select all)

☒ PERSON
 ☒ NORP
 ☐ FACILITY
 ☒ ORG

☒ GPE
 ☒ LOC
 ☒ PRODUCT
 ☐ EVENT

☐ WORK OF ART
 ☐ LANGUAGE
 ☒ DATE
 ☐ TIME

☐ PERCENT
 ☒ MONEY
 ☐ QUANTITY
 ☐ ORDINAL

☐ CARDINAL

Justin Pierre James Trudeau PERSON PC MP (born December 25, 1971 DATE) is a Canadian NORP politician serving as the 23rd prime minister of Canada GPE since 2015 DATE and Leader of the Liberal Party ORG since 2013 DATE .

Trudeau PERSON is the second-youngest Canadian NORP Prime Minister after Joe Clark PERSON ; he is also the first to be related to a previous holder of the post, as the eldest son of Pierre Trudeau PERSON .

Common textbook types:

Type	Examples
PERSON	Albert Einstein, Luca Maestri
ORG	Apple, the New York Times
GPE / LOC	Ulm, Germany, San Francisco
DATE / TIME	Tuesday, last year
MONEY	\$75 billion

Industry projects add types: drug names, SKUs, statute numbers, ticker symbols. The algorithm is the same idea. The label set is yours.

3. Gazetteers and rules

The simplest NER is a **gazetteer**: a list of client names, cities, and products you care about. Lookup is fast. If the list covers most of what you see, start there.

Gazetteers fail on new names, on aliases (*USA* vs *United States*), and on anything not in the file. You then need a process to update the list.

Rule-based NER uses token and POS patterns. A pattern like `NNP was born` says the proper noun is a person. Rules are precise on the cases you wrote and silent on the rest. They also break when the wording moves.

Use lists and rules as a baseline or as a fallback. Do not expect them to be the whole system once names are open-ended.

4. NER as sequence labeling

The practical approach is a model that, for each token, decides: is this an entity, and if so, of what type? That sounds like classification. The difference is **context**. A sentiment classifier can score a sentence without looking at the next one. NER cannot. If the previous token was `B-PERSON`, the current token is more likely `I-PERSON` (the last name) than a new organization.

This is **sequence labeling**, the same family as POS tagging. Standard encodings:

Tag	Meaning
<code>O</code>	Not an entity
<code>B-TYPE</code>	Beginning of a span of that type
<code>I-TYPE</code>	Inside / continuation of that span

Training data is token-level. Evaluation is usually span-level F1: the whole name and the type must match.

Features that used to be hand-built (shape, prefixes, surrounding POS) are now often learned by a bidirectional LSTM or a transformer. The modeling idea does not change: predict a tag path that is consistent left to right.

5. What this week is for

By the end of note **11.1** you should be able to:

- define NER and name the common entity types
- contrast a gazetteer, a rule list, and a sequence model
- explain why NER is sequence labeling and not document classification

Note **11.2** is what you do **after** you have the spans: linking and relations.

6. Practice

1. Why does a gazetteer struggle with *J. K. Rowling* if it only stores *Joanne Rowling*?
2. Tag `Ulm`, `Germany` with `B-` / `I-` / `O` if both tokens are a location. Then tag it if only `Ulm` is the location.
3. A movie-review sentiment model ignores neighboring sentences. Why is that usually fine for sentiment and usually wrong for NER?